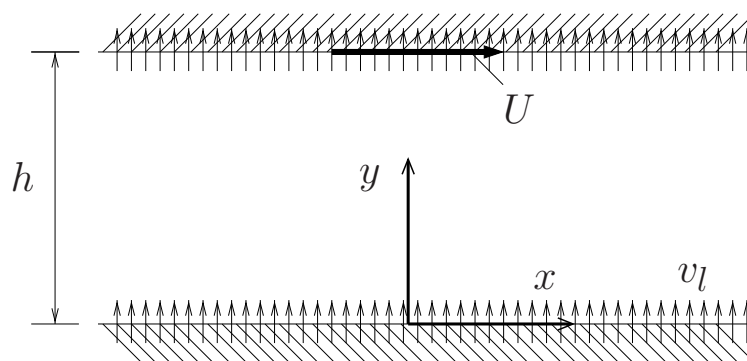


PROBLEM 2 (1 hour)

A perfect liquid of constant properties (density ρ , viscosity μ , thermal conductivity k and specific heat c) is confined between two infinite parallel plates at a distance h , as depicted in the figure below. The lower wall is at rest, while the upper one moves parallel to itself with a constant velocity U in the positive x -direction. The working fluid is injected through the lower porous wall into the channel with a uniform velocity v_l in the y -direction, while being removed from the channel through the upper porous plate with a uniform velocity v_u , also in the y -direction.



Gravitational forces can be neglected. The velocity field, $\vec{v} = u\vec{e}_x + v\vec{e}_y$, is assumed steady and independent of the x - and z -coordinates, $\partial/\partial t = \partial/\partial x = \partial/\partial z = 0$. The pressure at the upper wall has a constant value p_u . The lower and upper walls are kept at constant temperatures T_l and T_u , respectively. Known data: $(\rho, \mu, k, c, h, U, v_l, p_u, T_l, T_u)$.

1. (1 point). Obtain the only possible value of v_u consistent with the assumption of x -independence, and find $v(y)$.
2. (2.5 points). Determine the pressure field $p(x, y)$, and the horizontal velocity field, $u(y)$. Express $u(y)$ in terms of the Reynolds number based on the injection velocity, $Re = \rho v_l h / \mu$.
3. (1 point). Compute the volume flux per unit spanwise length, Q , across a surface perpendicular to the plates (z is the spanwise direction). Particularize for the limit $Re \ll 1$, and explain the result.
4. (1.5 points). Obtain the trajectory of the fluid particle with initial position $(x, y) = (0, x_0)$ at $t = 0$, and the corresponding pathline. Compute the time t_s at which the particle arrives at the upper wall, and the corresponding position x_s .
5. (1 point). Compute the viscous force per unit surface that the fluid exerts on the lower and upper walls. Particularize for the limit $Re \ll 1$, and explain the result.
6. (2 points). Write the differential equation and boundary conditions that determine the temperature field, $T(y)$. Obtain $T(y)$ neglecting the effect of viscous dissipation.
7. (1 point). Compute the conductive heat flux (heat transferred by conduction p.u. surface) from the fluid to each wall.
8. (2 additional points). Using an appropriately defined control volume, perform integral momentum and energy balances, and discuss the results of questions 5 and 7 in the light of these integral balances.

PROBLEM 2

$$\frac{\partial}{\partial x} = 0$$

$$(1) \nabla \cdot \vec{v} = 0 \Rightarrow \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \Rightarrow v = \text{CONSTANT} \Rightarrow \boxed{v_u = v_l = v}$$

$$(2) y\text{-MOMENTUM: } 0 = \frac{\partial p}{\partial y} \Rightarrow \boxed{p = p_u \forall x, y}$$

$$\begin{array}{l} \text{x-MOMENTUM: } \left[\rho v_l \frac{du}{dy} = \mu \frac{d^2 u}{dy^2} \right] \Rightarrow \boxed{u = U \frac{e^{\text{Re} \frac{y}{h}} - 1}{e^{\text{Re}} - 1}} \\ \hookrightarrow \left(\frac{\partial p}{\partial x} = 0 \right) \\ y=0: u=0 \\ y=h: u=U \end{array}$$

$$\text{TAYLOR EXPANSION OF } e^x \rightarrow e^x = 1 + x + \frac{x^2}{2} + \dots \text{ AS } x \rightarrow 0$$

$$\text{THIS, AS } \text{Re} \rightarrow 0, \quad u = U \frac{1 + \text{Re} \frac{y}{h} + O(\text{Re}^2) - 1}{1 + \text{Re} + O(\text{Re}^2) - 1} \Rightarrow \boxed{u = \frac{Uy}{h} \text{ AS } \text{Re} \rightarrow 0}$$

AND WE RECOVER COUETTE FLOW

$$(3) Q = \int_0^h u(y) dy = \frac{U}{e^{\text{Re}} - 1} \left[\frac{h}{\text{Re}} e^{\text{Re} \frac{y}{h}} - y \right]_0^h \Rightarrow \boxed{Q = \frac{Uh}{e^{\text{Re}} - 1} \left[\frac{1}{\text{Re}} (e^{\text{Re}} - 1) - 1 \right]}$$

$$\text{IN THE LIMIT } \text{Re} \rightarrow 0, \quad Q \rightarrow \frac{Uh}{2} \text{ (COUETTE)}$$

$$(4) \frac{dy}{dt} = v_l; \quad y(0) = 0 \rightarrow \boxed{y = v_l t} \rightarrow \boxed{t_s = \frac{h}{v_l}}$$

$$\frac{dx}{dt} = U \frac{e^{\text{Re} \frac{y}{h}} - 1}{e^{\text{Re}} - 1}; \quad x(0) = x_0 \rightarrow \int_{x_0}^x dx = \frac{U}{e^{\text{Re}} - 1} \int_0^t (e^{\text{Re} \frac{v_l t}{h}} - 1) dt$$

$$\Rightarrow \boxed{x - x_0 = \frac{U}{e^{\text{Re}} - 1} \left[\frac{h}{\text{Re} v_l} (e^{\text{Re} \frac{v_l t}{h}} - 1) - t \right]}$$

AND THEREFORE

$$\boxed{x_s = x_0 + \frac{Uh}{(e^{\text{Re}} - 1) v_l} \left(\frac{e^{\text{Re}} - 1}{\text{Re}} - 1 \right)}$$

$$(5) \begin{array}{l} \text{Diagram: } \vec{n}_u = -\vec{e}_y \text{ (top plate)}, \vec{n}_l = \vec{e}_y \text{ (bottom plate)} \\ \vec{f}_u = \vec{\tau}' \cdot (-\vec{e}_y) = -\mu \frac{du}{dy} \Big|_{y=h} \vec{e}_x = \frac{-\mu U \text{Re} e^{\text{Re}}}{h(e^{\text{Re}} - 1)} \vec{e}_x \\ \vec{f}_l = \vec{\tau}' \cdot (\vec{e}_y) = \mu \frac{du}{dy} \Big|_{y=0} \vec{e}_x = \frac{\mu U \text{Re}}{h(e^{\text{Re}} - 1)} \vec{e}_x \end{array}$$

AS $\text{Re} \rightarrow 0$

$$\begin{array}{l} \vec{f}_u \rightarrow -\frac{\mu U}{h} \vec{e}_x \\ \vec{f}_l \rightarrow \frac{\mu U}{h} \vec{e}_x \end{array} \quad \text{(COUETTE'S SOLUTION)}$$

$$\textcircled{6} \begin{cases} \rho c v_e \frac{dT}{dy} = \underbrace{\mu \left(\frac{du}{dy} \right)^2}_{\phi_0} + k \frac{d^2 T}{dy^2} \\ y=0: T=T_l \\ y=h: T=T_u \end{cases}$$

NEGLECTING ϕ_0 , $\frac{T-T_l}{T_u-T_l} = \frac{e^{\frac{Re Pr y}{h}} - 1}{e^{Re Pr} - 1}$

WHERE $Pr = \frac{\mu c}{k}$

$$\textcircled{7} \quad \vec{n}_u = \vec{e}_y \quad q_u = \vec{q} \cdot \vec{n} = -k \vec{n} \cdot \nabla T$$

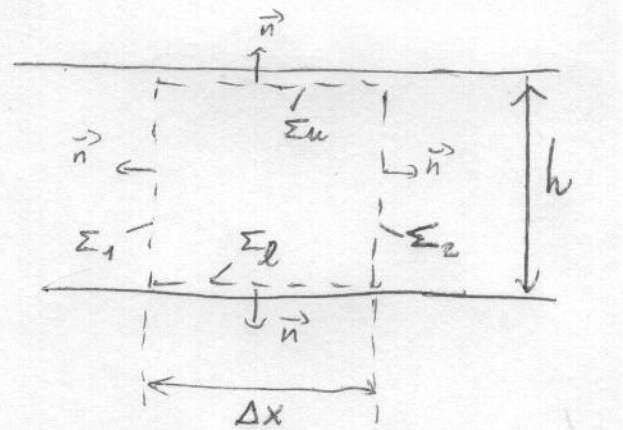
UPPER WALL: $q_u = -k \frac{dT}{dy} \Big|_{y=h} = -\rho c v_e (T_u - T_l) \frac{e^{Re Pr} - 1}{e^{Re Pr} - 1}$

LOWER WALL: $q_l = k \frac{dT}{dy} \Big|_{y=0} = \rho c v_e (T_u - T_l) \frac{1}{e^{Re Pr} - 1}$

$\textcircled{8}$ MOMENTUM EQUATION:

$$\int_{\Sigma_u, \Sigma_l} \rho \vec{v} (\vec{v} \cdot \vec{n}) d\sigma = -p_u \int_{\Sigma_c} \vec{n} d\sigma + \int_{\Sigma_1, \Sigma_2} \vec{\tau} \cdot \vec{n} d\sigma$$

$\frac{\partial}{\partial x} = 0$ $\frac{\partial}{\partial x} = 0$



$$\Rightarrow \rho v_e (v_e \vec{e}_x + v_e \vec{e}_y) \Delta x - \rho v_e^2 \vec{e}_y \Delta x =$$

$$= [\tau'_{xy}(y=h) - \tau'_{xy}(y=0)] \Delta x \vec{e}_x \Rightarrow$$

$$\Sigma_c = \Sigma_1 \cup \Sigma_2 \cup \Sigma_l \cup \Sigma_u$$

$$\Rightarrow \rho v_e v = \mu \left(\frac{du}{dy} \Big|_{y=h} - \frac{du}{dy} \Big|_{y=0} \right)$$

IN AGREEMENT WITH $\textcircled{5}$
(DIFFERENCE IN SHEAR STRESSES CAUSED BY NET MOMENTUM FLUX IN THE X-DIRECTION)

INTERNAL ENERGY EQUATION (NEGLECTING DISSIPATION!)

$$\int_{\Sigma_1, \Sigma_2} \rho e (\vec{v} \cdot \vec{n}) d\sigma = - \int_{\Sigma_l, \Sigma_u} \vec{q} \cdot \vec{n} d\sigma \Rightarrow q_u + q_l = \rho c v_e (T_u - T_l)$$

$\frac{\partial}{\partial x} = 0$ $\frac{\partial}{\partial x} = 0$

IN AGREEMENT WITH $\textcircled{7}$

CONDUCTIVE HEAT FLUX BALANCES NET INTERNAL ENERGY FLUX THROUGH THE CONTROL VOLUME